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Matematika Sains Data

Ujian Tengah Semester
Semester Genap TA 2025/2026

Dosen Pengampu:
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- Parameter Personalisasi

N (2 digit terakhir NIM) → NIM mod 100 → N = 15
a (digit puluhan NIM) → floor (N/10) → floor (15/10) → 1
b (digit satuan NIM) → N mod 10 → b = 5
K (konstanta skala data) → (a+b+1) → 1+5+1 → K = 7
θ_0 (sudut awal bilangan kompleks) → (a . 30) + (b . 5) → θ_0 = (1 . 30) + (5 . 5) = 30 + 25 = 55
α (learning rate gradient descent) → 0,001 * (b+1) → α = 0,001 . 6 = 0,006

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3. Segmentasi Pelanggan dan Komposisi Fungsi Harga

↳ himpunan pelanggan

↳ $A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\}$ → media sosial

$= \{15, 17, 19, 22, 25, 28, 30\}$ → total : 7

↳ $B = \{N+3, N+5, N+7, N+10, N+12, N+18\}$ → referral

$= \{18, 20, 22, 25, 27, 33\}$ → total: 6

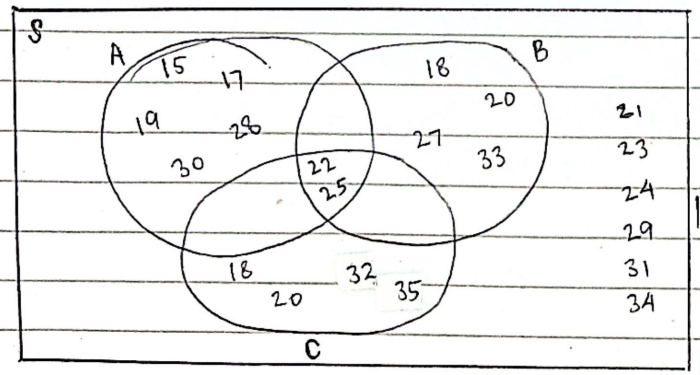
↳ $C = \{N+1, N+7, N+10, N+11, N+17, N+20\}$ → loyalty program

$= \{16, 22, 25, 26, 32, 35\}$ → total : 6

↳ $U = \{N, N+1, N+2, \ldots, N+20\}$ (total: 21 pelanggan)

$= \{15, 16, 17, \ldots, 35\}$

a. menggambar diagram Venn



$|A \cup B \cup C| = \{15, 16, 17, 18, 19, 20, 22, 25, 26, 27, 28, 30, 32, 33, 35\}$

$|A \cap B \cap C| = \{22, 25\}$

$|(A \cup B)| = \{15, 17, 18, 19, 20, 22, 25, 27, 28, 30, 33\}$

$|C'| = \{15, 17, 18, 19, 20, 21, 23, 24, 27, 28, 29, 30, 31, 33, 34\}$

$|(A \cup B) \cap C'| = \{15, 17, 18, 19, 20, 27, 28, 30, 33\}$

$|A \oplus B| = \{15, 17, 18, 19, 20, 27, 28, 30, 33\}$

c. fungsi $f(x)$ dan $g(x)$.

↳ $f(x) = K.x + (a+b) = 7x + (1+5) = 7x + 6$

↳ $g(x) = (x^2)/(b+1) + a = \frac{x^2}{5+1} + 1 = \frac{x^2}{6} + 1$

↳ $(f \circ g)(x) = f(g(x)) = 7(g(x)) + 6 = 7\left(\frac{x^2}{6} + 1\right) + 6 = \frac{7x^2}{6} + 7 + 6 = \frac{7x^2}{6} + 13$

$(f \circ g)(x) = \frac{7}{6}x^2 + 13$

↳ $(g \circ f)(x) = (g(f(x)) = \frac{1}{6}(f(x))^2 + 1 = \frac{1}{6}(7x+6)^2 + 1 = \frac{1}{6}(49x^2 + 84x + 36) + 1$

$= \frac{49}{6}x^2 + 14x + 6 + 1 = 49/6\, x^2 + 14x + 7.$

Dipindai dengan CamScanner

↳ Apakah komposisi komutatif? Buktikan dengan contoh numerik $x = b+2$.

$x = b+2 \rightarrow x = 5+2 = 7$

↳ $(f \circ g)(7) = \frac{7}{6}(7)^2 + 13 = \frac{7}{6}(49) + 13 = 343/6 + 13$

↳ $(g \circ f)(7) = 49/6\,(7)^2 + 14(7) + 7 = 49^2/6 + 98 + 7 = 49^2/6 + 105$

↳ Komposisi tidak komutatif karena $(f \circ g)(7) \neq (g \circ f)(7)$

d. menghitung $f^{-1}(x)$

↳ $f(x) = y$, $y = 7x+6 \rightarrow y-6 = 7x \rightarrow$ 7 dipindahkan kekiri $= (y-6)/7 = x$

$f^{-1}(x) = \frac{x-6}{7}$

↳ menghitung volume pembelian x yang dihasilkan

pendapatan $= (K.100 + a + b)$ ribu $= (7.100 + 1 + 5) = 700 + 6 = 706$ ribu

$x = f^{-1}(706) = \frac{706-6}{7} = \frac{700}{7} = 100$

↳ Volume pembelian x yang dihasilkan adalah 100 unit.

e. Klasifikasi fungsi f dan g

↳ $f(x) = 7x+6$

↳ injektif : fungsi disebut injektif bila $f(x_1) = f(x_2)$

$f(x_1) = f(x_2) \rightarrow 7x_2 + 6 = 7x_1 + 6$

maka fungsi $f(x)$ injektif.

↳ Surjektif : setiap $y \in R$, terdapat $x = \frac{y-6}{7}$ merupakan bilangan riil.

maka fungsi $f(x)$ surjektif.

↳ bijektif : $f(x)$ bersifat injektif dan surjektif, maka $f(x)$ bijektif.

↳ $g(x) = \frac{x^2}{6} + 1$

↳ injektif : misal $x_1 = 2$, $x_2 = -2$ $f(x_1) = f(x_2)$ tetapi $x_1 \neq x_2$

$f(2) = \frac{4}{6} + 1$, $f(-2) = \frac{4}{6} + 1$ maka $g(x)$ tidak injektif.

↳ surjektif : nilai di bawah 1 tidak masuk dalam range, maka $g(x)$ tidak.

↳ bijektif : $g(x)$ bukan bijektif karena bukan injektif maupun surjektif.

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